The Health Informer: RAG Model Comparison for Health/Wellness Inquiries

Team 13: Health AI - Ashhad Jaffer and Naseem Heydari

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1. Problem Statement

Health is foundational to every aspect of our lives, and as we learn more about our bodies, we inevitably generate questions about our health. We inquire across a large range of topics that are relevant for our general wellness, such as exercise, nutrition, sleep, mental health, and more. Doctors and other professionals are not always readily available for these everyday questions, so most people turn to search engines first. But consumer health websites often are too generic, forums can spread misinformation, and the peer-reviewed medical literature that could actually provide reliable answers is written for researchers, not the general public. More recently, people have turned to Large Language Models (LLMs) for quick answers. However, standard LLMs generate responses without grounding them in verifiable sources, making them prone to hallucination, which is particularly dangerous when the topic is someone's health. The key for us to get credible answers for our questions is to bridge the gap between the jargon of medical literature and the accessibility of LLMs. Our answer for that is HealthInformer.

HealthInformer is a Retrieval-Augmented Generation (RAG) system designed to retrieve its answers from credible PubMed medical literature and provide those insights into plain language for users. A user types a health question in plain language, maybe asking how many hours of sleep one should get, and the system retrieves the relevant sleep passages from a curated corpus of PubMed abstracts. With that context , the LLM creates a plain language output that is easy to understand for the user, but is still grounded in peer-reviewed evidence, along with citations for each claim. HealthInformer is designed to help users understand entry-level health and wellness questions over a myriad of topics; it is not, however, a diagnostic tool or a replacement for an actual medical professional.

Our corpus spans 21,926 PubMed abstracts across 36 health topic categories, from cardiovascular health, diabetes, nutrition, mental health, and sexual health. All are chunked into 40,096 passages and indexed using BioMedBERT embeddings, a transformer model pre-trained on biomedical literature. We compare two LLM backends (Claude 3.5 Haiku and Llama 3.3 70B) across multiple evaluation dimensions, including the RAGAS (Automated Evaluation of Retrieval Augmented Generation) framework (Es et al., 2024) for RAG-specific metrics and the PubMedQA benchmark (Jin et al., 2019), a dataset of expert-annotated biomedical questions used to measure answer accuracy against a 78% human performance ceiling. In our report, we'll discuss where the system succeeds, where it falls short, and what we learned about the challenges of making retrieval work when user language and scientific language don't match.

2. Methodology

Building HealthInformer required decisions across three layers: what data to use, how to retrieve the right pieces of it, and how to generate answers that are both easy to understand for users and faithful to the peer-reviewed literature. Each layer had its own challenges, and some of our most important design decisions came from actual failures we encountered along the way.

- Data Collection and Corpus Design

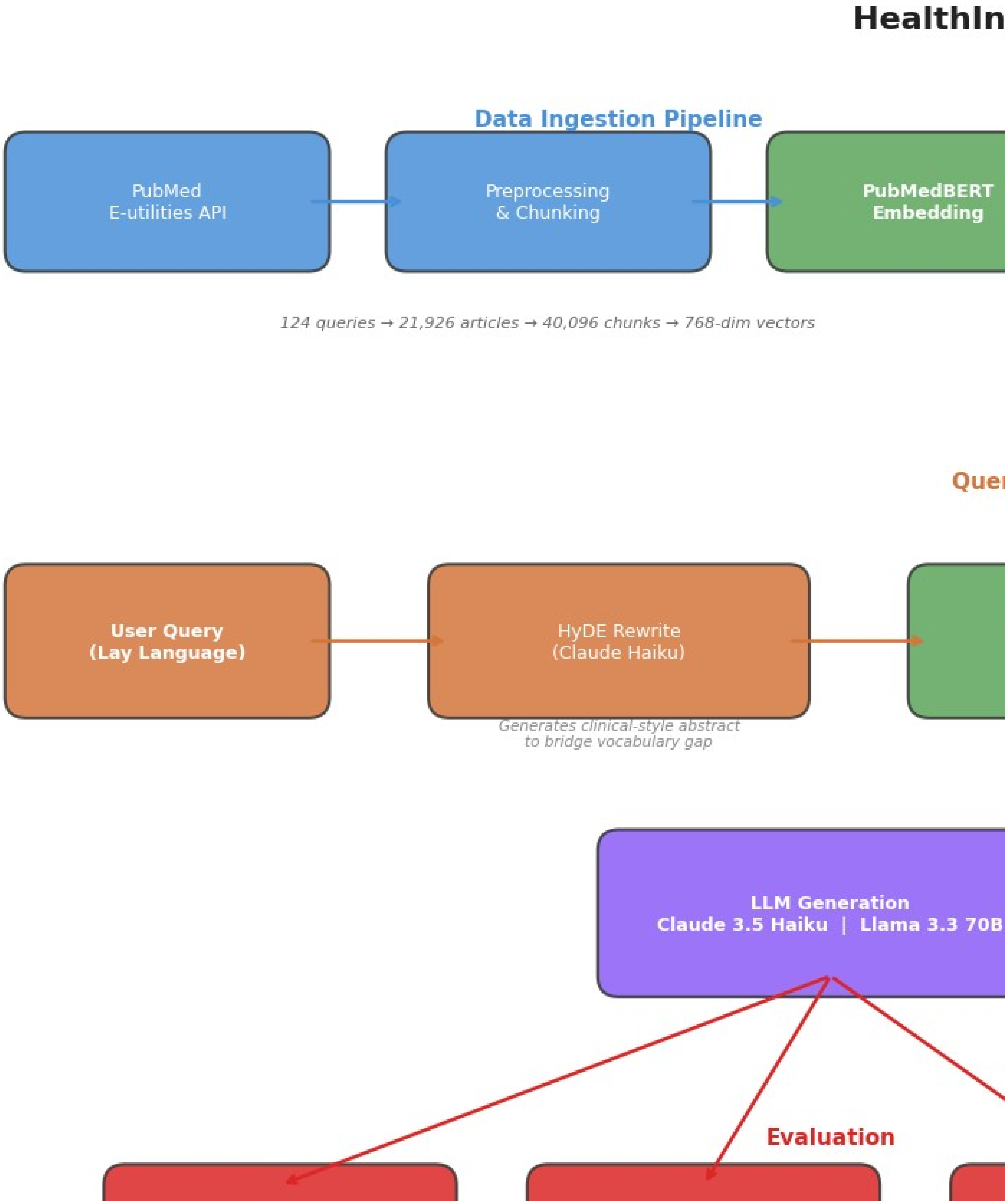
We sourced our medical literature from PubMed, the National Library of Medicine's database of biomedical research. Using PubMed's E-utilities API, we retrieved abstracts across 124 search queries involving 36 health topic categories, from broad areas like cardiovascular health and mental health to more specific wellness topics like sleep hygiene, oral health, and substance use. With PubMed, our corpus contained 21,926 unique articles for those topics. We split each abstract into smaller passages at sentence boundaries, targeting a maximum of 512 tokens per chunk to match BioMedBERT's context window. Each chunk overlaps with the next by one sentence so that context isn't lost at the boundaries. This produced 40,096 chunks from our 21,926 articles. Each chunk carries its source metadata (PubMed ID, title, authors, journal, and a direct link) so every claim in an outputted answer can be traced back to its original source.

- **Retrieval Pipeline**

For embedding our chunks, we used BioMedBERT, a transformer model that is pre-trained specifically on medical text. Compared to general embedding models, BioMedBERT is able to better capture the understanding of medical language. These embeddings are stored in ChromaDB, a lightweight vector database that we run locally and handles similarity search over our 40,096 chunks. When a user asks a question, the system embeds it, finds the eight (K=8) most similar chunks by similarity search, and passes those as context to the LLM.

When we started testing our pipeline, we ran into an issue: BioMedBERT was trained on abstracts, not on the kind of casual questions a real user would type. A question like "what are the warning signs of diabetes?" outputted an embedding that was too far from the clinical language in PubMed abstracts, and the retriever returned papers on completely unrelated topics like urinary tract infections. So it wasn’t a problem with the LLM, the evidence it received deviated from the actual topic of the query, meaning the answer would deviate as well

To address this, we tested three retrieval strategies to fix: a keyword-based query rewrite, a prose-style clinical rewrite, and Hypothetical Document Embeddings (HyDE) (Gao et al., 2023). HyDE works by first asking the LLM to generate a short, hypothetical abstract answering the user's question, and then embedding that hypothetical passage instead of the actual query (Gao et al., 2023). Because the generated text uses clinical language and structure similar to real PubMed abstracts, it is much closer to relevant documents in the embedding space. HyDE improved relevant chunk retrieval from 28% to 66%, the single most impactful change we made to the system. Our unified pipeline is shown below in Figure 1:


HealthIn
Data Ingestion Pipeline
PubMed
E-utilities API
Preprocessing
& Chunking
PubMedBERT
Embedding
124 queries → 21,926 articles → 40,096 chunks → 768-dim vectors
Quer
User Query
(Lay Language)
HyDE Rewrite
(Claude Haiku)
Generates clinical-style abstract
to bridge vocabulary gap
LLM Generation
Claude 3.5 Haiku | Llama 3.3 70B
Evaluation

- **Generation and Model Comparison**

Alongside the working chatbot, we also wanted to compare two LLM backends for HealthInformer, both hosted on AWS Bedrock: Claude 3.5 Haiku, a commercial model from Anthropic, and Llama 3.3 70B, an open-source model from Meta.

Both models receive the same inputs: the user's original question, the eight retrieved context chunks, and a system prompt instructing the model to write in plain language, cite each claim with a numbered reference to a specific PubMed article, and avoid giving personal medical advice. The pipeline is built as a custom RAG chain, with LangChain used only for the RAGAS evaluation.

- **User Interface**

The frontend is a simple Streamlit chat application. Users type a health question and receive a plain-language response with inline numbered citations. Below each answer, expandable source cards show the title, authors, journal, and a direct link to each cited PubMed article so users can verify claims for themselves. There’s also a disclaimer banner at the top of the interface that states HealthInformer is an educational tool and not a diagnosis tool or replacement for professional medical advice.

3. Evaluation Strategy

We evaluated HealthInformer using three approaches: an automated RAG evaluation framework (RAGAS), a standardized medical QA benchmark (sources from PubMed), and a manual spot check of generated answers.

Our primary evaluation tool was RAGAS (Retrieval Augmented Generation Assessment), a framework designed specifically for evaluating RAG pipelines. RAGAS measures four metrics: faithfulness (whether the LLM's claims are actually supported by the retrieved chunks), answer relevancy (whether the response addresses the user's question), context precision (whether the retrieved chunks are relevant and well-ranked), and context recall (whether the retriever captured the information needed to fully answer the question). Each metric produces a score between 0 and 1. We ran RAGAS on 53 curated test questions spanning our 36 topic categories, with both models evaluated under the same conditions.

We made an important methodological decision when choosing which LLM to use as the RAGAS evaluator. We realized that if the same model generates the answers and then judges them, the evaluation becomes circular. To avoid this, we used a fresh instance of Claude 3.5 Haiku as the evaluator for both models (not the same instance of Haiku that's participating in the comparison). This means Haiku evaluates its own outputs (an unavoidable limitation) but also serves as a neutral judge for Llama, which keeps the comparison fair.

For our second evaluation approach, we benchmarked both models against PubMedQA (Jin et al., 2019), a dataset of 1000 expert-annotated biomedical questions where each answer is yes, no, or maybe (we benchmarked 500 from it). PubMedQA provides its own gold-standard context passages rather than using our retriever, so this measures the LLM's medical reasoning ability in isolation. The human performance ceiling on this benchmark is 78%.

Finally, we manually reviewed 50 generated responses per model, scoring each on a 1–5 quality scale for clarity, completeness, and appropriateness for a general audience. We also flagged any hallucinations (claims not supported by the retrieved sources).

4. Results

Before we could evaluate our LLMs and extract the results, we had to fix the retrieval pipeline. Our initial results showed that BioMedBERT struggled to match casual user questions with clinical abstract language, only retrieving relevant chunks 28% of the time. After implementing HyDE, that number jumped to 66%. The impact on the RAGAS metrics was big, as shown in Table 1:

Metric	Pre-HyDE	Post-HyDE	Change
Faithfulness	0.743	0.898	0.155
Answer Relevancy	0.68	0.914	0.234
Context Precision	0.213	0.474	0.261
Context Recall	0.296	0.56	0.264

***Table 1**

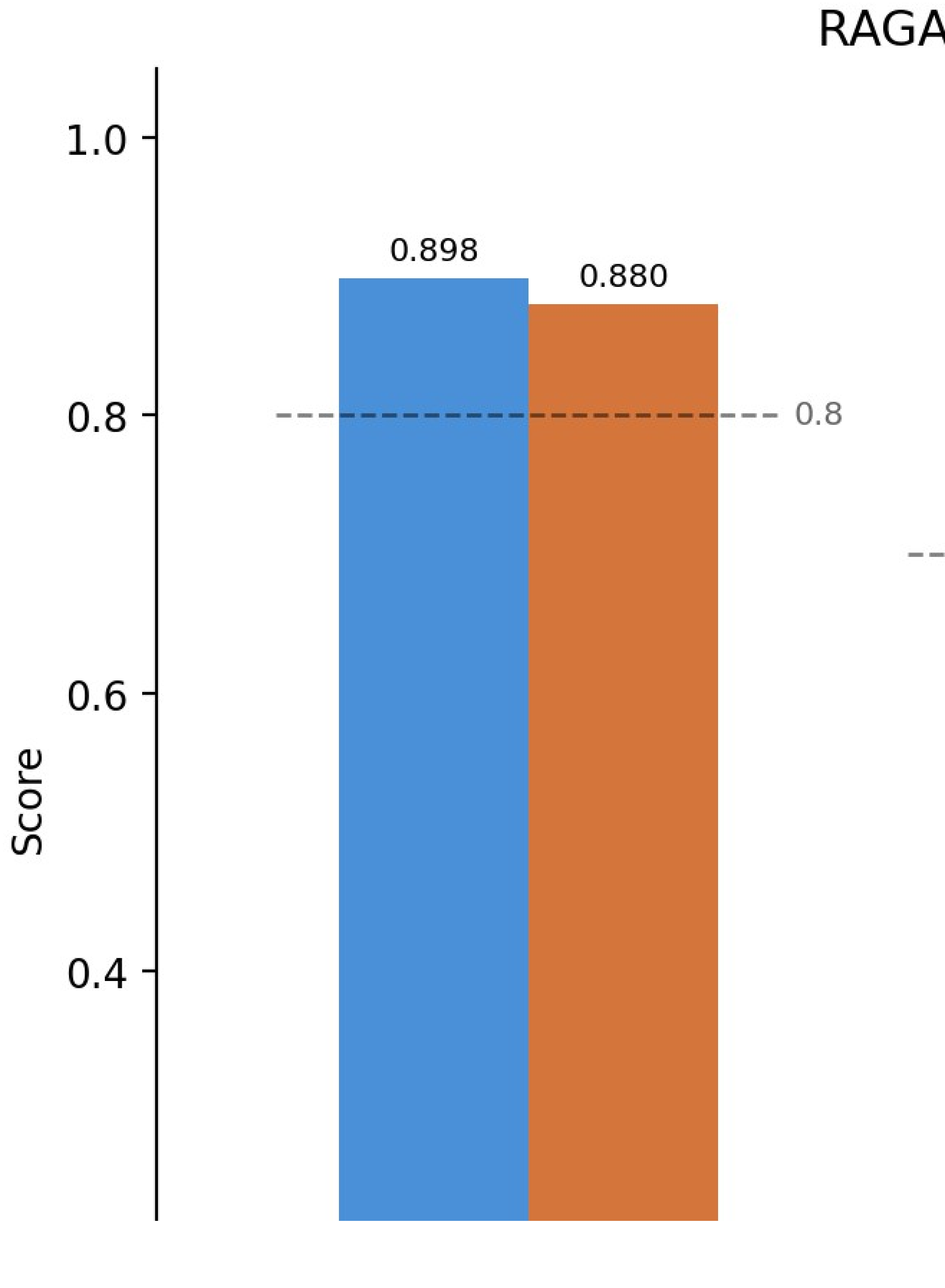
Two of four RAGAS metrics flipped from failing to passing after HyDE, and all four improved substantially. This confirmed retrieval was the bottleneck, not generation. With HyDE in place, we ran the full evaluation across both models displayed in Table 2:

Metric	Claude 3.5 Haiku	Llama 3.3 70B
Faithfulness	0.898	0.88
Answer Relevancy	0.914	0.898
Context Precision	0.474	0.47
Context Recall	0.56	0.57
PubMedQA Accuracy	65.60%	77.20%
Quality (manual)	4.28 / 5	3.96 / 5
Hallucinations	0 / 50	0 / 50
Avg Latency	13.0s	5.5s

***Table 2**

Both models scored high on faithfulness and answer relevancy, indicating that LLMs generate responses that are grounded in the retrieved evidence and relevant to the user's question. Neither model hallucinated in our manual spot check review. The generation side of the pipeline is working well.

Figure 2 below shows the RAGAS metric comparison between both models. Faithfulness and answer relevancy are strong for both, while context precision and recall lag behind, validating our hypothesis that the retrieval layer, not the generation layer, is the system's limitation.



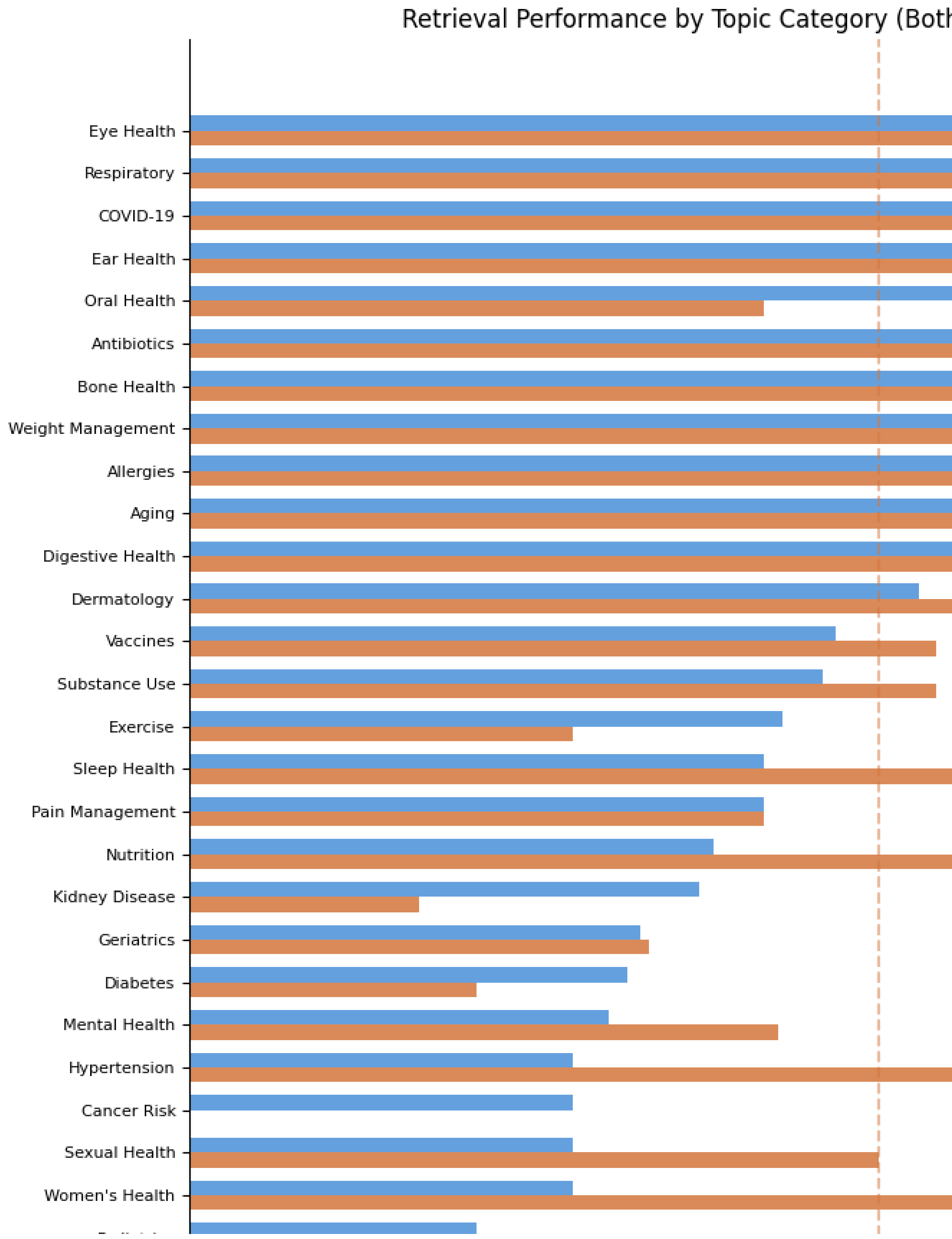
RAGA
1.0
0.8
0.6
0.4
Score
0.898
0.880
0.8

***Figure 2**

Context precision and recall remain the weakest metrics for both models, sitting below our original targets (0.7 for Precision, 0.6 for Recall). We hypothesize that this is a retrieval limitation, not a generation one. Even with HyDE, the retriever still uses some irrelevant chunks, particularly for the broader wellness topics where the vernacular gap between user language and clinical abstracts is wider.

The most interesting finding is the gap between models for the PubMedQA. Llama scored 77.2%, approaching the 78% human ceiling, while Haiku scored 65.6%. Hallucination isn’t the reason for this gap (because neither model hallucinates). Rather Haiku tends to default to "maybe" even on questions where the evidence supports a clear yes or no, predicting "maybe" 161 times versus only 66 genuinely ambiguous items in the dataset. In a medical context, this over-caution could be seen as a feature rather than a flaw, but the fact that it still has instances where it errs to caution for definitive topics is concerning. On the flip side, Llama rarely hedges on the items ind that data set. Haiku does produce more polished, readable responses, reflected in its higher manual spot check quality score of 4.28 versus Llama's 3.96.

We also study the retrieval performance by topic category, as shown below in Figure 3. Topics with direct clinical terminology in PubMed, like vaccines and diabetes, show stronger retrieval scores. Broader wellness topics like exercise and nutrition show weaker performance, where the vocabulary gap between the casual user queries and clinical abstracts is harder for HyDE to fully bridge.



Retrieval Performance by Topic Category (Both
Eye Health
Respiratory
COVID-19
Ear Health
Oral Health
Antibiotics
Bone Health
Weight Management
Allergies
Aging
Digestive Health
Dermatology
Vaccines
Substance Use
Exercise
Sleep Health
Pain Management
Nutrition
Kidney Disease
Geriatrics
Diabetes
Mental Health
Hypertension
Cancer Risk
Sexual Health
Women's Health

***Figure 3**

Neither model seems to be objectively better. Haiku writes cleaner answers and stays more cautious; Llama is faster, more decisive on benchmark questions, and equally faithful to its sources. The choice between them depends on whether the user prioritizes response quality or throughput.

5. Broader Impacts

We designed HealthInformer to make medical literature more accessible, but we realize that any tool that puts health information in front of non-experts carries real risks that need to be acknowledged.

The biggest risk is over-trust by the user. Even with a disclaimer banner, users may treat HealthInformer's responses, or any other similar tool's, as medical advice rather than a point for understanding. The system is grounded in peer-reviewed evidence, but it retrieves and summarizes; it doesn't diagnose, and it can't account for an user's full medical history. We try to mitigate this through disclaimers and by instructing the LLM to avoid giving personal medical recommendations, but we can't control a user's actions based on the response.

Hallucination is common in LLMs, and in a medical context, it's especially dangerous. Our evaluation found a 0% hallucination rate across 100 manually reviewed responses, and both models scored around 0.88 on RAGAS faithfulness, meaning claims are well-supported by retrieved sources (which is what we aimed for). However, zero hallucinations in a sample of 100 doesn’t guarantee zero hallucinations at scale. RAG reduces this risk significantly compared to a standard LLM by grounding every response in retrieved evidence, but we can’t say there is no risk of hallucination.

There is also bias in the underlying literature itself. PubMed articles disproportionately represent certain demographics. For instance, many clinical studies are concentrated toward Western populations, and conditions affecting underrepresented groups may not have as much coverage in our corpus. HealthInformer persists in whatever gaps exist in the research it retrieves. This as a known limitation when we chose the source material.

Finally, privacy is a topic that always needs to be addressed. We made sure that HealthInformer doesn’t require a login, doesn’t store queries beyond the active session, and doesn’t log personal data.

6. Conclusion / Future Work

We wanted to answer one question with HealthInformer: can a RAG pipeline produce accurately cited health information that is both grounded in real research and accessible to everyday users? If it can, then the general public can use this to increase their everyday knowledge about their health and make more informed choices. Based on our evaluation, the answer is a yes, but there are important caveats for that answer, especially concerning the system's limitations.

Both LLMs performed well on the generation side. Faithfulness scores above 0.88, zero hallucinations across 100 manually reviewed responses, and strong answer relevancy scores show that when given the right evidence, both Haiku and Llama produce reliable, readable answers. The challenge is getting them the right evidence in the first place. Context precision and recall were our weakest metrics throughout the project, and the HyDE fix, while helpful, did not fully close that gap. The primary difficulty is a vocabulary mismatch: users ask questions in plain language, but the medical literature is written in clinical terminology. Bridging that gap at the retrieval layer still remains the primary problem.

If we were to continue developing HealthInformer, the most impactful next step would be thinking about a cross-encoder reranker to improve context precision. It would re-score the retrieved chunks with a more powerful model before passing them to the LLM. Other improvements we thought of are: expanding the corpus to include full-text articles from PubMed Central's open-access subset, running user studies to measure whether real users actually find the answers helpful, and tracking product-level metrics like user satisfaction and conversation depth. We also think there is value in testing how the system handles questions that fall outside its corpus, since we found in our PubMedQA evaluation that Haiku tends to refuse rather than hallucinate when evidence is ambiguous, which is a promising safety behavior worth studying further. All of these points aren’t really in the scope of this project but it’s definitely ideas worth exploring.

HealthInformer is not a finished product. But as a proof of concept, it shows that a RAG pipeline can make peer-reviewed medical evidence meaningfully more accessible for users and their everyday health queries.

7. Statement of Work

Ashhad Jaffer: Lead Developer, co-writer, co-manual checker of LLM outputs

Naseem Heydari: Lead Writer, co-developer, co-manual checker of LLM outputs

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